Industrial Documentation Standards and Frameworks

1. System Requirements: ISO/IEC/IEEE 29148 (Superseding IEEE 830)

Structure the introduction and requirements section based on the **Software Requirements Specification (SRS)** standard.

- **Note:** Students must separate user requirements from system capabilities and explicitly list business rules.
- **Report content:**
    - **Functional Requirements:** Specifically, data-centric actions (e.g., *"The system must allow a Clerk to update inventory counts"*).
    - **Non-Functional Requirements:** Database-specific constraints like security (encryption at rest), availability, or performance (e.g., *"Search queries must return results in under 200ms"*).

2. Conceptual & Logical Modeling: ISO/IEC 19505 (UML) or IE Crow's Foot

Industry standard tools (like MySQL Workbench, draw.io, Erwin, or Lucidchart) default to **Information Engineering (IE) Crow's Foot Notation** or **UML Class Diagrams**.

- **Note:** Students need to read and produce diagrams that software engineers and enterprise architects actually use.
- **What to do:** Students must choose between:
    - **IE Crow's Foot Notation:** For standard relational modeling.
    - **UML Class Diagrams (ISO/IEC 19505):** If students are integrating the database with object-oriented backends (Java/Python).
    - *Note:* If students want to use Elmasri's EER notation, please provide a **Mapping Appendix** converting it to an industry-standard Physical Diagram.

3. Data Dictionary: ISO/IEC 11179 (Metadata Standards)

Students should include a formalized **Data Dictionary** based on the core principles of **ISO/IEC 11179** for metadata registries.

Students are required to have a standardized Markdown or Word table for every table in their schema. For example,

Attribute Name	Data Type	Key Type	Nullable?	Default Value	Business Rules / Constraints
emp_id	VARCHAR(10)	PK	No	None	Must start with 'EMP-' followed by 6 digits.
salary	DECIMAL(10,2)	None	No	0.00	Must be greater than or equal to minimum wage.
dept_id	INT	FK	Yes	NULL	References DEPARTMENT(dept_id) on delete SET NULL.

4. SQL Code Quality: SQLStyle.guide (Simon Holywell) / Google SQL Style Guide

Students must follow an open industry style guide, such as **SQLStyle.guide** or the **Google SQL Style Guide** to prepare SQL scripts which will be handed to database administrators (DBAs)

- **Note:** Clean coding habits. SQL code must be nicely written and readable.
- **Core rules to enforce from these standards:**
  - Use meaningful, lowercase snake_case for identifiers (e.g., customer_orders, not tblCustOrders or CustomerOrders).
  - Explicitly capitalize SQL keywords (SELECT, INSERT, JOIN, WHERE).
  - Proper indentation for joins and subqueries.